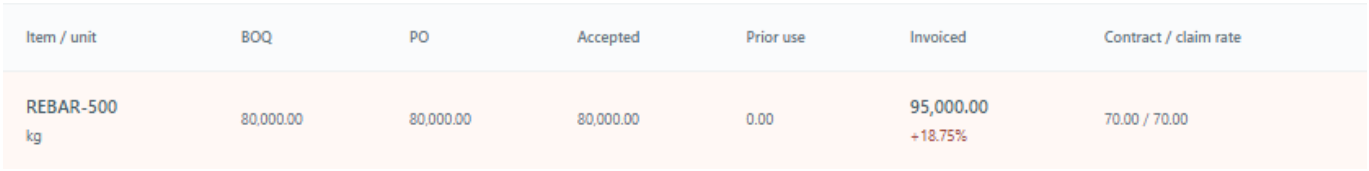
BuildTrust AI: a decision you can explain

Construction finance with evidence at every step.

A payment request should answer a simple question: what supports this amount? BuildTrust AI puts the approved scope, purchase order, accepted delivery and supplier invoice side by side. It checks the claim and the cash, then asks an independent person to approve the exact payment.

The application is implemented. Its security hardening and recorded checks are complete for the local judge package. It uses fictional projects and simulated settlement. No bank, card or real money is connected.



Item / unit	BOQ	PO	Accepted	Prior use	Invoiced	Contract / claim rate
REBAR-500 kg	80,000.00	80,000.00	80,000.00	0.00	95,000.00 +18.75%	70.00 / 70.00

Actual invoice comparison showing the claimed quantity and the supporting records

Three examples you can trace

Example	What happens in the app
An inflated invoice	95 tonnes claimed, 80 supported. The app identifies BDT 1,050,000 excess and blocks payment.
An eligible payment	A different person approves BDT 70,000. Cash changes from BDT 1,000,000 to 930,000 once. A repeated execution returns the same receipt.
A changed material rate	Twenty tonnes at BDT 74,000 instead of 70,000 increases the estimate by BDT 80,000. Forecast cash changes from 600,000 to 520,000. Recorded cash remains 2,000,000.

These are tested fictional examples. Their source records, outputs and audit references are included in the execution evidence.

One workspace for the project team

The 14 application areas connect document review with quantity surveying, prices, estimates, contracts, accounting, forecasts and payment evidence. Operators prepare work. Finance Managers check it. Admins approve eligible actions independently. Auditors read the records and verify the local audit chain.

An estimate approval creates a budget baseline. It does not change an existing purchase order or authorize a payment. Forecasts keep recorded cash separate from expected receipts. Missing cost evidence and dates remain visible.

Security is implemented

Private Admin and Finance accounts require an authenticator. The public demo keeps four fixed README passwords and does not require MFA. Its database and files are separate from the private workspace.

The backend checks current sessions, roles and project access. It recalculates financial decisions from saved records. Upload validation, safe browser requests, outbound-provider restrictions, secret scanning, usage caps and private backup/restore have recorded acceptance evidence.

Executed check	Result
Local backend suite	251 passed
Frontend tests	10 passed
Private browser workflows	20 passed
Public demo browser logins	Four passed without MFA
Payment HTTP scenarios	A-H all passed
Hosted Windows and Linux	Both passed. Each skipped one optional OCR test because its language files were not installed.

The October 20 hosted run identifies tested application revision 0009a198bf9ce7583cd2c870953e21d82a584739. Later documentation changes retain the same application behavior. The publication receipt beside the source archive records the final package and its matching checks.

The 19-control security checklist records 17 verified controls. Native database row-level security is deferred while SQLite is used. Incoming webhooks are disabled, so signature verification is not applicable. A local database owner can rewrite the audit chain. These checks do not replace an independent security assessment or establish public-deployment readiness.

A quick route through the submission

1. Read this overview and watch the 4:18 demonstration.
2. Open the nine-slide presentation and its speaker notes.
3. Try the separate public demo using the command and four passwords in the README.
4. Read the technical and security reports for the rules behind each screen.
5. Inspect the execution evidence and source archive to trace the results.

Repository: BuildTrust AI on GitHub. The README explains installation. The feature guide shows the full workflow. Remaining work names the external reviews, optional service checks and human submission actions still needed.